SIMATS ENGINEERING

Engineer to Excel

SIMATS

Saveetha Institute of Medical And Technical Sciences

(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CO1 - ASSESSMENT TOOL 2

COURSE NAME: DESIGN AND ANALYSIS OF ALGORITHMS

COURSE CODE: CSA06

COURSE OUTCOME COVERED

CO1: Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method.

(BL-1, BL-2, BL-3)

Assessment Tool Weightage:

QUESTIONS: 10 **Total Marks: 100**

Annexure A

Concept Mapping

Code of Conduct:

I P.Sai Ganesh Reddy(192525111) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

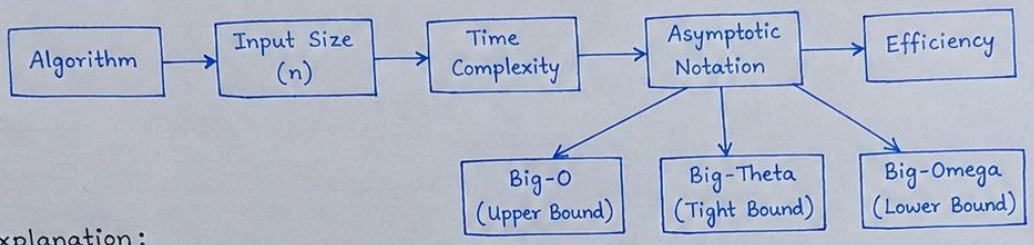
P.Sai Ganesh Reddy

RUBRICS FOR CALCULATION

Criteria	**Weigh tage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

1. Consider the concept map: Algorithm → Input Size (n) → Time Complexity → Asymptotic Notation → Efficiency. Explain the directional relationship between each node and show how an algorithm is evaluated as the input size increases. Extend the map by adding the branches Big-O, Big-Theta and Big-Omega under Asymptotic Notation. Justify how these notations help classify the growth of running time for large values of n. Interpret how the completed map supports the measurement of algorithm efficiency in a structured way.

Concept Map:



Explanation:

1. Algorithm → Input Size (n):
   - The input size (n) represents the amount of data given to the algorithm. As n increases, the running time of the algorithm changes.
2. Input Size (n) → Time Complexity:
   - The time taken by the algorithm depends on the input size. This relationship is called time complexity.
3. Time Complexity → Asymptotic Notation:
   - The running time is expressed using asymptotic notations (Big-O, Big-Theta, Big-Omega) to describe how it grows with n, especially for large values.
4. Asymptotic Notation → Efficiency:
   - These notations help us compare and classify algorithms, and finally we can measure the efficiency of an algorithm.

Role of Big-O, Big-Theta and Big-Omega:

- Big-O (O): gives an upper bound on the growth rate (worst-case).
- Big-Theta (Θ): gives a tight bound (exact order of growth).
- Big-Omega (Ω): gives a lower bound (best-case).

Together, these notations help classify the growth of running time for large values of n, because they show whether the time grows slowly, moderately or rapidly.

Interpretation:

The completed map shows a clear flow from the algorithm to its efficiency through input size, time complexity and asymptotic notation. It gives a structured way to evaluate an algorithm by understanding how the running time grows and which bound it falls under. This helps in comparing different algorithms and choosing the most efficient one for large inputs.